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These solutions are not the only “perfect” or “10/10” solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Recall, from the Week 4 Tutorial Activity: if $U, W \subseteq V$ are subspaces such that every vector in $\mathbf{v} \in U + W$ can be written uniquely $\mathbf{w} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$ then we say that the sum $U + W$ is direct and write $U + W = U \oplus W$.

Q1. Suppose that V is finite dimensional and let $T : V \rightarrow V$ be a linear map such that $T^2 = T$.

(a) Prove that $V = \text{Image}(T) \oplus \ker(T)$.

Solution:

1. Firstly, let us show that $\text{Image}(T) + \ker(T) = V$. To do this, fix an arbitrary vector $\mathbf{v} \in V$ and express $\mathbf{v} = T(\mathbf{v}) + [\mathbf{v} - T(\mathbf{v})]$. Clearly, $T(\mathbf{v}) \in \text{Image}(T)$, so we only need to prove that $\mathbf{v} - T(\mathbf{v}) \in \ker(T)$. Note that

$$T(\mathbf{v} - T(\mathbf{v})) = T(\mathbf{v}) - T(T(\mathbf{v})) = T(\mathbf{v}) - T^2(\mathbf{v}) = T(\mathbf{v}) - T(\mathbf{v}) = \mathbf{0}.$$

2. Now, to prove that the sum is direct, we need to check that $\text{Image}(T) \cap \ker(T) = \{\mathbf{0}\}$. To do this, fix an arbitrary vector $\mathbf{v} \in \text{Image}(T) \cap \ker(T)$. Since $\mathbf{v} \in \text{Image}(T)$, there exists some $\mathbf{u} \in V$ such that $\mathbf{v} = T(\mathbf{u})$. Also, since $\mathbf{v} \in \ker(T)$, $T(\mathbf{v}) = \mathbf{0}$. Thus,

$$\mathbf{0} = T(\mathbf{v}) = T(T(\mathbf{u})) = T^2(\mathbf{u}) = T(\mathbf{u}) = \mathbf{v}.$$

Q2. Suppose that W_1 and W_2 are subspaces of V with $\dim(W_1) = n$, $\dim(W_2) = k$, and $n \leq k$. Prove the following:

(a) $\dim(W_1 \cap W_2) \leq n$

Solution: Since $W_1 \cap W_2 \subseteq W_1$, $\dim(W_1 \cap W_2) \leq \dim(W_1) = n$.

(b) $\dim(W_1 + W_2) \leq n + k$

Solution: Let α be a basis for W_1 , and let β be a basis for W_2 so that $|\alpha| = n$ and $|\beta| = k$. Since $\alpha \cup \beta$ is a spanning set for $W_1 + W_2$, we can conclude that $\dim(W_1 + W_2) \leq |\alpha \cup \beta| \leq n + k$.

Provide examples of W_1 and $W_2 \subseteq \mathbb{R}^3$ such that $n < k$ and:

(a) $\dim(W_1 \cap W_2) = n$

Solution: If $W_1 = \{\mathbf{0}\}$ and W_2 is a line, then $\dim(W_1 \cap W_2) = \dim(\{\mathbf{0}\}) = 0 = \dim(W_1)$.

(b) $\dim(W_1 + W_2) = n + k$

Solution: Using the example above, $\dim(W_1 + W_2) = \dim(W_2) = \dim(W_1) + \dim(W_2)$.

Q3. Suppose that $T : V \rightarrow V$ is a linear transformation.

(a) Prove that $S(\mathbf{v}) = T(T(\mathbf{v}))$ is also a linear transformation.

Solution: For every $\mathbf{u}, \mathbf{v} \in V$ and $k \in \mathbb{R}$, we have that

$$S(\mathbf{u} + k\mathbf{v}) = T(T(\mathbf{u} + k\mathbf{v})) = T(T(\mathbf{u}) + kT(\mathbf{v})) = T(T(\mathbf{u})) + kT(T(\mathbf{v})) = S(\mathbf{u}) + kS(\mathbf{v}),$$

so S is linear.

(b) Give an example of a non-zero transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(T(\mathbf{v})) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^2$.

Solution: Consider, for example, the linear transformation given by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

The transformation T is non-zero, since $T(0, 1) = (1, 0)$. Nevertheless,

$$T \left(T \begin{pmatrix} x \\ y \end{pmatrix} \right) = T \begin{pmatrix} y \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

for all $x, y \in \mathbb{R}$.

(c) If $T(T(\mathbf{v})) = \mathbf{0}$ for all $\mathbf{v} \in V$, what can you conclude about $\text{image}(T)$ and $\ker(T)$?

Solution: We can conclude that $\text{Image}(T) \subseteq \ker(T)$.

Q4. Given a set of vectors $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ consider the coordinate linear map $T : \mathbb{R}^n \rightarrow V$ given by

$$T(x_1, \dots, x_n) = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n.$$

(a) What statement about T corresponds to the statement $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$?

Solution: We will prove that S is a spanning set for V if, and only if, T is surjective:

($\Rightarrow$) If T is surjective, that means that for every vector $\mathbf{v} \in V$, there is a tuple of real numbers $(x_1, \dots, x_n) \in \mathbb{R}^n$ such that $T(x_1, \dots, x_n) = \mathbf{v}$. This means, for every vector $\mathbf{v} \in V$, there are real numbers $x_1, \dots, x_n \in \mathbb{R}$ such that $\mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n$, so $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

($\Leftarrow$) If S is a spanning set for V , then for every $\mathbf{v} \in V$ there are real numbers $x_1, \dots, x_n \in \mathbb{R}$ such that $\mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n$. This means that for every vector $\mathbf{v} \in V$, there is a tuple of real numbers $(x_1, \dots, x_n) \in \mathbb{R}^n$ such that $T(x_1, \dots, x_n) = \mathbf{v}$, so T is onto.

(b) What statement about T corresponds to the statement $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent?

Solution: We will prove that S is linearly independent if, and only if, T is injective:

($\Rightarrow$) If T is injective, that means that the only tuple of real numbers $(x_1, \dots, x_n) \in \mathbb{R}^n$ such that $T(x_1, \dots, x_n) = \mathbf{0}$ is the tuple $(0, \dots, 0)$. This means, for all real numbers $x_1, \dots, x_n \in \mathbb{R}$, if $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n = \mathbf{0}$, then $x_1 = \dots = x_n = 0$, so S is linearly independent.

($\Leftarrow$) If S linearly independent, then $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n = \mathbf{0}$ implies that $x_1 = \cdots = x_n = 0$. This means that for every tuple of real numbers $(x_1, \dots, x_n) \in \mathbb{R}^n$, if $T(x_1, \dots, x_n) = \mathbf{0}$, then $(x_1, \dots, x_n) = (0, \dots, 0)$. So, $\ker(T) = \{\mathbf{0}\}$, and therefore T is one-to-one.

Q5. Suppose that W_1 and W_2 are subspaces of a finite-dimensional vector space V . Prove that: If

$$\dim(W_1) + \dim(W_2) > \dim(V)$$

then there is a non-zero vector $\mathbf{v} \in W_1 \cap W_2$.

Solution: Let $\alpha = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis for W_1 , and let $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ be a basis for W_2 . If it is the case that the set $\alpha \cup \beta$ is linearly independent, then all of its elements have to be different, and that would give us $(\dim(W_1) + \dim(W_2))$ -many linearly independent vectors in V , which is impossible. Thus, the set $\alpha \cup \beta$ is linearly dependent, and there are real numbers $a_1, \dots, a_n, b_1, \dots, b_m$, not all of them zero, such that

$$a_1\mathbf{u}_1 + \cdots + a_n\mathbf{u}_n + b_1\mathbf{w}_1 + \cdots + b_m\mathbf{w}_m = \mathbf{0}.$$

Since α is linearly independent, it cannot be the case that all the b_j 's are 0. Also, since β is linearly independent, it cannot be the case that all the a_i 's are zero either. Thus, the vector

$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_n\mathbf{u}_n = -b_1\mathbf{w}_1 - \cdots - b_m\mathbf{w}_m$$

is a non-zero vector which is in W_1 (being a linear combination of the $\mathbf{u}_i$'s) and in W_2 (being a linear combination of the $\mathbf{w}_j$'s).

Q6. Suppose that V and W are finite-dimensional vector spaces. Prove the following:

(a) There exists an injective linear transformation $T : V \rightarrow W$ if, and only if, $\dim(W) \geq \dim(V)$.

Solution: We proved during the lecture that if there exists an injective linear transformation $T : V \rightarrow W$, then $\dim(V) \leq \dim(W)$ as a consequence of the Dimension Theorem.

Assume now that $\dim(V) \leq \dim(W)$, and fix bases $\alpha = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ for V and $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ for W so that $m \leq n$. Consider the linear transformation given by $T(\mathbf{v}_i) = \mathbf{w}_i$ for every $i \leq m$. To check that T is injective, let $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m$ be a vector in V such that $T(\mathbf{v}) = \mathbf{0}$. Then

$$\mathbf{0} = T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \cdots + a_mT(\mathbf{v}_m) = a_1\mathbf{w}_1 + \cdots + a_m\mathbf{w}_m.$$

Since β is linearly independent, $a_1 = \cdots = a_m = 0$, $\mathbf{v} = \mathbf{0}$, and T is one-to-one.

(b) There exists a surjective linear transformation $T : V \rightarrow W$ if and only if $\dim(W) \leq \dim(V)$.

Solution: We proved during the lecture that if there exists a surjective linear transformation $T : V \rightarrow W$, then $\dim(V) \geq \dim(W)$ as a consequence of the Dimension Theorem.

Assume now that $\dim(V) \geq \dim(W)$, and fix bases $\alpha = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ for V and $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ for W so that $m \geq n$. Consider the linear transformation given by

$$T(\mathbf{v}_i) = \begin{cases} \mathbf{w}_i & i \leq n \\ \mathbf{0} & i > n \end{cases}$$

To check that T is surjective, let $\mathbf{w} = a_1\mathbf{w}_1 + \cdots + a_n\mathbf{w}_n$ be a vector in W and consider the vector $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ in V . Then

$$\mathbf{w} = a_1\mathbf{w}_1 + \cdots + a_n\mathbf{w}_n = a_1T(\mathbf{v}_1) + \cdots + a_nT(\mathbf{v}_n) = T(a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n) = T(\mathbf{v}),$$

so T is onto.

Q7. Suppose that $\dim(V) = k$. Prove the following:

- (a) Any k linearly independent vectors form a basis of V .

Solution: Let S be a linearly independent set with k -many elements. If S is not a basis, then there exists some vector $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span}(S)$. This implies that $S \cup \{\mathbf{v}\}$ is linearly independent, and therefore $\dim(V) \geq k + 1$, which is a contradiction.

- (b) Any k vectors that span V form a basis of V .

Solution: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a spanning set for V with k -many elements. If S is not a basis, then there exists a non-trivial, null linear combination

$$a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = \mathbf{0}.$$

Since some $a_i \neq 0$, then we can express the linear dependence above as

$$\mathbf{v}_i = -\frac{a_1}{a_i} \mathbf{v}_1 - \dots - \frac{a_{i-1}}{a_i} \mathbf{v}_{i-1} - \frac{a_{i+1}}{a_i} \mathbf{v}_{i+1} - \dots - \frac{a_k}{a_i} \mathbf{v}_k,$$

and therefore $\mathbf{v}_i \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k\}$, which implies that $\dim(V) \leq k - 1$; a contradiction.